

14.0 Milestone Assessment: Network Communication Protocols

The ultimate objective of network technology is **communication**. Network communication is organized by protocols, or sets of rules and standards that define how data is formatted, transmitted, received, and processed across a computer network. These protocols allow different hardware devices, operating systems, and applications to communicate and exchange data reliably and securely.

Network protocols perform specific functions at different layers of the **OSI model**, and multiple protocols work together to provide end-to-end communication.

Data Link Layer — Layer 2

At the Data Link layer, Ethernet and Address Resolution Protocol (ARP) support communication across the local network.

- **Ethernet (IEEE 802.3)** provides framing and MAC addressing for communication on Ethernet networks. Ethernet frames include a Frame Check Sequence (FCS), which uses a Cyclic Redundancy Check (CRC) to detect transmission errors.
- **ARP (Address Resolution Protocol)** resolves an IPv4 address to the corresponding MAC address on the local network. For example, before a computer can send an Ethernet frame to its IPv4 default gateway, ARP can be used to determine the gateway's MAC address.

Network Layer — Layer 3

At the Network layer, Internet Protocol (IP) and Internet Control Message Protocol (ICMP) support logical addressing, routing, and network diagnostics.

- **IP (Internet Protocol)** provides logical source and destination addressing and allows packets to be routed between networks using IPv4 or IPv6.
- **ICMP (Internet Control Message Protocol)** carries control, error, and diagnostic information. For example, the **ping** utility uses ICMP Echo Request and Echo Reply messages to test whether another device can be reached.

Transport Layer — Layer 4

TCP and UDP provide transport services between applications, but they are designed for different communication requirements.

- **TCP (Transmission Control Protocol)** is connection-oriented and provides reliable, ordered delivery. TCP establishes a connection using a three-way handshake, divides application data into segments, uses sequence numbers and acknowledgments, and retransmits missing data when necessary.
- **UDP (User Datagram Protocol)** is connectionless and has less overhead because it does not establish a connection or provide TCP-style acknowledgments, sequencing, and retransmission.

TCP is appropriate when reliable and ordered delivery is more important than minimizing latency. Examples include HTTP web traffic, SSH remote administration, RDP sessions, SMB file sharing, SMTP email, and FTP file transfers.

UDP is appropriate when low overhead and timely delivery are more important than retransmitting every lost packet. Examples include many real-time voice and video applications, DNS queries, DHCP, and NTP.

The choice therefore depends on the requirements of the application. A file transfer should not silently lose part of a file, making TCP appropriate. During a real-time voice call, however, retransmitting an old voice packet may be less useful than continuing to deliver current audio, making UDP appropriate for many real-time media applications.

Application Layer — Layer 7

Application-layer protocols provide network services used by applications and users.

- **DNS (Domain Name System)** resolves domain names to IP addresses and provides other DNS records. A client typically sends a DNS query using UDP port 53, while TCP port 53 is also used when required.
- **DHCP (Dynamic Host Configuration Protocol)** automatically provides clients with network configuration such as an IP address, subnet mask, default gateway, DNS servers, and lease information. DHCP uses UDP ports 67 and 68.
- **HTTP (Hypertext Transfer Protocol)** allows web clients and servers to exchange web content and commonly uses TCP port 80. Secure web communication uses HTTPS with TLS, normally over TCP port 443.
- **FTP (File Transfer Protocol)** transfers files between clients and servers using TCP port 21 for the control connection and TCP port 20 for active-mode data transfer. Because traditional FTP does not encrypt its traffic, secure alternatives such as SFTP or FTPS should be considered when confidentiality is required.
- **SMTP (Simple Mail Transfer Protocol)** is used to send and relay email. SMTP commonly uses TCP port 25 for server-to-server transport and TCP port 587 for message submission.
- **SSH (Secure Shell)** provides encrypted remote command-line administration and secure communications over TCP port 22. For example, a network administrator can use SSH instead of Telnet to securely configure a router or switch.

- **RDP (Remote Desktop Protocol)** provides remote graphical access to another computer and commonly uses TCP and UDP port 3389. A help-desk technician could use RDP to remotely access an authorized Windows workstation for troubleshooting.
- **SMB (Server Message Block)** provides shared access to files, printers, and other network resources, commonly using TCP port 445. For example, an employee accessing a shared Windows network folder may be using SMB.
- **NTP (Network Time Protocol)** synchronizes clocks between network devices and commonly uses UDP port 123. Accurate time is important for logs, authentication, security investigations, and troubleshooting.
- **SIP (Session Initiation Protocol)** establishes, manages, and terminates multimedia communication sessions such as VoIP calls. SIP commonly uses ports 5060 and 5061, with 5061 commonly associated with SIP over TLS.
- **SNMP (Simple Network Management Protocol)** is used to monitor and manage network devices such as routers, switches, access points, and printers. For example, a network management system can use SNMP to collect device status and performance information.

How the Protocols Work Together

A single network communication normally requires several protocols working together across the OSI model.

For example, when a user enters a website name into a browser, DNS can resolve the domain name to an IP address. HTTP or HTTPS provides the application communication. TCP can establish a reliable transport connection between the client and web server. IP supplies the logical source and destination addresses and allows packets to be routed between networks. On the local Ethernet network, ARP may resolve the IPv4 address of the next-hop device to its MAC address, and Ethernet encapsulates the data into frames for local delivery.

The receiving system processes the communication in the opposite direction, removing the Ethernet, IP, and TCP information and ultimately delivering the application data to the appropriate service.

This demonstrates that protocols do not operate independently. Each protocol performs a specialized function, and together they provide the addressing, delivery, reliability, security, and application services required for devices to communicate across modern networks.



THE MOST COMMON NETWORK COMMUNICATION PROTOCOLS



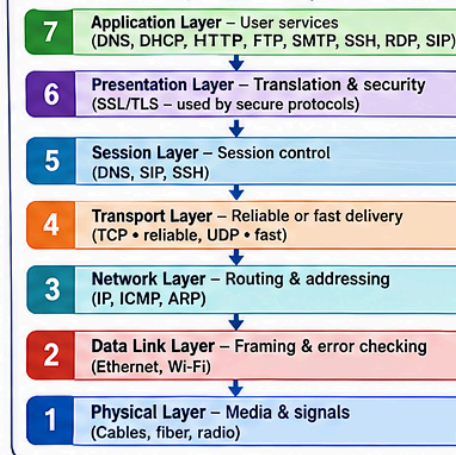
How Data Moves Across Networks and the Internet

#	PROTOCOL (Layer)	PURPOSE / WHAT IT DOES	HOW IT WORKS (In Simple Terms)	COMMON PORT(S)	USES / EXAMPLES
1	DNS (Application)	 Translates domain names to IP addresses and other network info.	A client queries a DNS server for a domain name. The server returns the corresponding IP address (or other DNS record).	53 (UDP/TCP)	- Website name resolution - Mail routing (MX records) - Service discovery
2	DHCP (Application)	 Automatically assigns IP addresses and other network configuration settings to devices.	A device sends a DHCP request to a server, which responds with an IP address, subnet mask, gateway, DNS server, and lease time.	67 (Server/UDP) 68 (Client/UDP)	- IP address assignment - Network configuration - Lease management
3	RDP (Presentation)	 Provides remote access to a computer's desktop and resources.	Runs over TCP to allow a user to connect to and control a remote system's desktop.	3389 (TCP)	- Remote administration - Help desk support - Access to remote systems
4	NTP (Application)	 Synchronizes clocks of devices across a network.	Clients query an NTP server to get the correct time and adjust their clocks for accuracy.	123 (UDP)	- Time synchronization - Logging and auditing - Authentication (Kerberos)
5	SSH (Application)	 Provides secure encrypted access to network devices or servers.	Creates an encrypted session (SSH) over TCP for remote command line access and secure file transfer.	22 (TCP)	- Remote server access - Secure file transfers - Network device mgmt.
6	SMB (Application)	 Allows shared access to files, printers, and other resources on Windows networks.	Enables clients to read/write files and communicate with servers over the network.	445 (TCP)	- File and printer sharing - Windows file services - Inter-process comm.
7	SIP (Application)	 Manages sessions for voice, video, and multimedia calls.	Establishes, manages, and ends media sessions between participants.	5060 (UDP/TCP) 5061 (TCP/TLS)	- VoIP call setup - Video conferencing - Instant messaging
8	ARP (Datalink)	 Maps IP addresses to MAC addresses on a local network.	Broadcasts a request for the MAC address for an IP, then caches the result.	— (Layer 2)	- Local network communication - Default gateway lookup - Device discovery
9	HTTP (Application)	 Transfers web pages and content between a client and a server.	Uses TCP (port 80) to request and deliver web content.	80 (TCP)	- Web browsing - REST APIs - Web applications
10	FTP (Application)	 Transfers files between a client and a server over a network.	Uses TCP for reliable file transfer.	21 (Control TCP) 20 (Data TCP)	- Upload/download files - File sharing
11	SMTP (Application)	 Sends email from a mail client to a mail server and between mail servers.	Uses TCP on port 25 to deliver email messages.	25, 587 (TCP)	- Email sending - Mail server communication
12	IP (Network)	 Addresses and routes packets across networks.	Specifies source and destination IP addresses and moves packets between networks.	— (Layer 3)	- Internet communication - Packet delivery
13	ICMP (Network)	 Reports errors and network conditions.	Sends control messages (e.g., ping, unreachable) via IP.	— (Layer 3)	- Troubleshooting (ping) - Network diagnostics
14	UDP (Transport)	 Fast, lightweight communication without connection setup.	Sends data without reliability checks (no handshaking).	53, 67, 123, 161, 5004 (UDP)	- Streaming media - Online gaming - VoIP
15	TCP (Transport)	 Reliable, ordered, error-checked data delivery.	Establishes a connection, sends data in order, and checks for errors.	80, 443, 21, 25 (TCP)	- Web browsing (HTTPS) - File transfer (FTP) - Reliable applications

TCP vs UDP: KEY DIFFERENCES

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable delivery (acknowledgments & retransmission)	Best-effort (no guarantee)
Ordering	In order	No order
Speed	Slower	Faster
Overhead	Higher	Lower
Use Cases	Web, email, file transfer	Streaming, VoIP, Gaming, DNS

HOW PROTOCOLS WORK TOGETHER (OSI MODEL)



<https://herbal-rancher.github.io/portfolio/>